

Report Sensitivity Model 3

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0.0.1 1. Discrepancies between the reported and implemented analysis

The paper states that the dependent variable is log-transformed after adding a small constant (0.05) to accommodate zero trauma shares. In the original estimation code we examined, we do not find a '+0.05' transformation step. Accordingly, reproducing the published results does not require (and is not consistent with) a $\log(y + 0.05)$ specification.

Instead, the published estimates are exactly reproduced when the outcome is constructed as follows: the log is taken only for strictly positive trauma shares, and observations with a trauma share of zero are not transformed into a finite log value but are treated as left-censored at a fixed censoring point on the log scale (approximately -3). This is a substantively different procedure from adding a constant before logging. Under the implemented approach, zeros contribute to the likelihood through a censoring assumption ("latent below the threshold") rather than through an observed transformed value. Because this censoring step is central to the model's interpretation and to how zeros are handled, the mismatch between the paper's description and the implemented code is consequential.

0.0.2 2. Why the censored Tobit/intreg approach is conceptually misaligned with the data

The Tobit specification can be seen as a mechanical attempt to solve the "log of zero" problem. However, this choice inadvertently imposes structural assumptions that are conceptually misaligned

with bioarchaeological data. The trauma measure is a proportion derived from counts—trauma cases observed out of the number of examined skeletons in a site–period. In this setting, a zero share typically means “0 cases out of n examined,” not that an underlying continuous latent variable has been censored by a detection limit. Tobit’s key assumption—that there exists an unbounded latent normal outcome that can fall below a threshold and then becomes censored—does not naturally correspond to a “count out of n ” data-generating process.

A related concern is that the information content of a zero depends heavily on the denominator. Observing 0 cases out of a large n provides much stronger evidence that the underlying rate is near zero than observing 0 cases out of a small n . In the authors’ Tobit setup, zeros are handled via a fixed censoring point (roughly equivalent to a 5% violence rate). In this model, both 0/5 and 0/500 enter as “left-censored below the same threshold”. Weights can change influence, but the likelihood still does not reflect the binomial sampling evidence that makes 0/500 far more informative than 0/5.

Finally, proportional/count outcomes have a mean–variance relationship dictated by binomial sampling (and often additional overdispersion). The Tobit framework assumes (on the latent scale) a constant-variance normal disturbance; analytic weights can rescale contributions to the likelihood, but they do not make the Tobit likelihood equivalent to the count-based sampling process that generated the data, nor do they automatically recover the correct dependence of variance on both n and the mean.

0.0.3 3. Why a beta-binomial model is conceptually better

A beta-binomial model matches the structure of the data directly. It models the observed number of trauma cases k out of n examined individuals, with the probability of trauma π linked to covariates (e.g., via a logit link). The binomial component handles the fact that the data are discrete counts bounded by n ; the beta component allows π to vary across site–periods beyond what a strict binomial would permit, capturing overdispersion that is plausible in aggregated bioarchaeological data.

This approach resolves the key conceptual issues above:

- Zeros are treated as ordinary outcomes with an exact probability $P(k = 0)$, rather than as censored latent values requiring an **arbitrary censor threshold**.
- Differences in denominators are handled natively through the likelihood: 0/500 and 0/10 are not treated as equivalent signals.
- Predicted means remain within $[0, 1]$ by construction.
- Overdispersion can be modeled explicitly, which generally yields more credible uncertainty estimates than relying on a homoskedastic latent-normal assumption.

0.1 4. Empirical Analysis and Comparison

0.1.1 4.1 Sensitivity of the Tobit results to the censoring point

The Tobit specification requires choosing a censoring point c on the log scale (the paper’s implementation corresponds to approximately $c \approx \log(0.05) \approx -3.0$). Because this threshold is not implied by the underlying “count out of n ” data-generating process, we re-estimated the full Tobit model across a grid of plausible censoring values $c \in \{-5.0, -4.5, -4.0, -3.5, -3.0, -2.5\}$, holding the covariates, analytic weights, clustering, and estimation procedure fixed (see **Figure 1** and **Table 1**).

Coefficient instability and sensitivity of statistical support. The figure and coefficient grid show that several effects shift as c changes. For example, `Early_Bronze_Age` is not statistically significant at very low censor points ($c = -5.0, -4.5$) but becomes significant for higher thresholds (from $c = -4.0$ onward). Conversely, `Region1` is statistically supported at lower censor points ($c = -5.0, -4.5, -4.0$) but loses support as the threshold rises (from roughly $c = -3.5$ onward). Substantively, key context coefficients such as `Battle_Sites` and `Royal_Interments` also shrink as c increases, indicating that effect magnitudes are not stable to this modeling choice.

Fit vs. calibration trade-off. Within the Tobit likelihood, fit improves mechanically as c increases (logLik rises and AIC/BIC fall). However, this improvement coincides with sharply worsening calibration of zeros: while the observed weighted censoring rate is **0.125**, the model’s predicted weighted censoring probability rises from **0.126** at $c = -5.0$ to **0.600** at $c = -3.0$ and **0.711** at $c = -2.5$. Thus, the censor points that appear “best” by AIC/BIC correspond to the most implausible predicted frequency of zeros and exacerbate the model’s misalignment with the bounded outcome.

Table 1: Model Fit versus Calibration. Table 1: Model Fit versus Calibration.

cpoint	logLik	AIC	BIC	obs_cens_w	pred_cens_w
-5.0	-124.957	275.914	307.201	0.125	0.126
-4.5	-119.917	265.834	297.121	0.125	0.159
-4.0	-114.720	255.439	286.727	0.125	0.258
-3.5	-109.478	244.957	276.244	0.125	0.441
-3.0	-104.356	234.713	266.000	0.125	0.600
-2.5	-99.551	225.101	256.389	0.125	0.711

Note: Each row reports the Tobit/intreg-style model re-estimated with an alternative censoring point (c) (log scale), holding covariates, analytic weights, clustering, and estimation identical. (`obs_cens_w`) is the observed weighted share of left-censored observations (zeros), and (`pred_cens_w`) is the model-implied weighted probability of censoring averaged over observations.

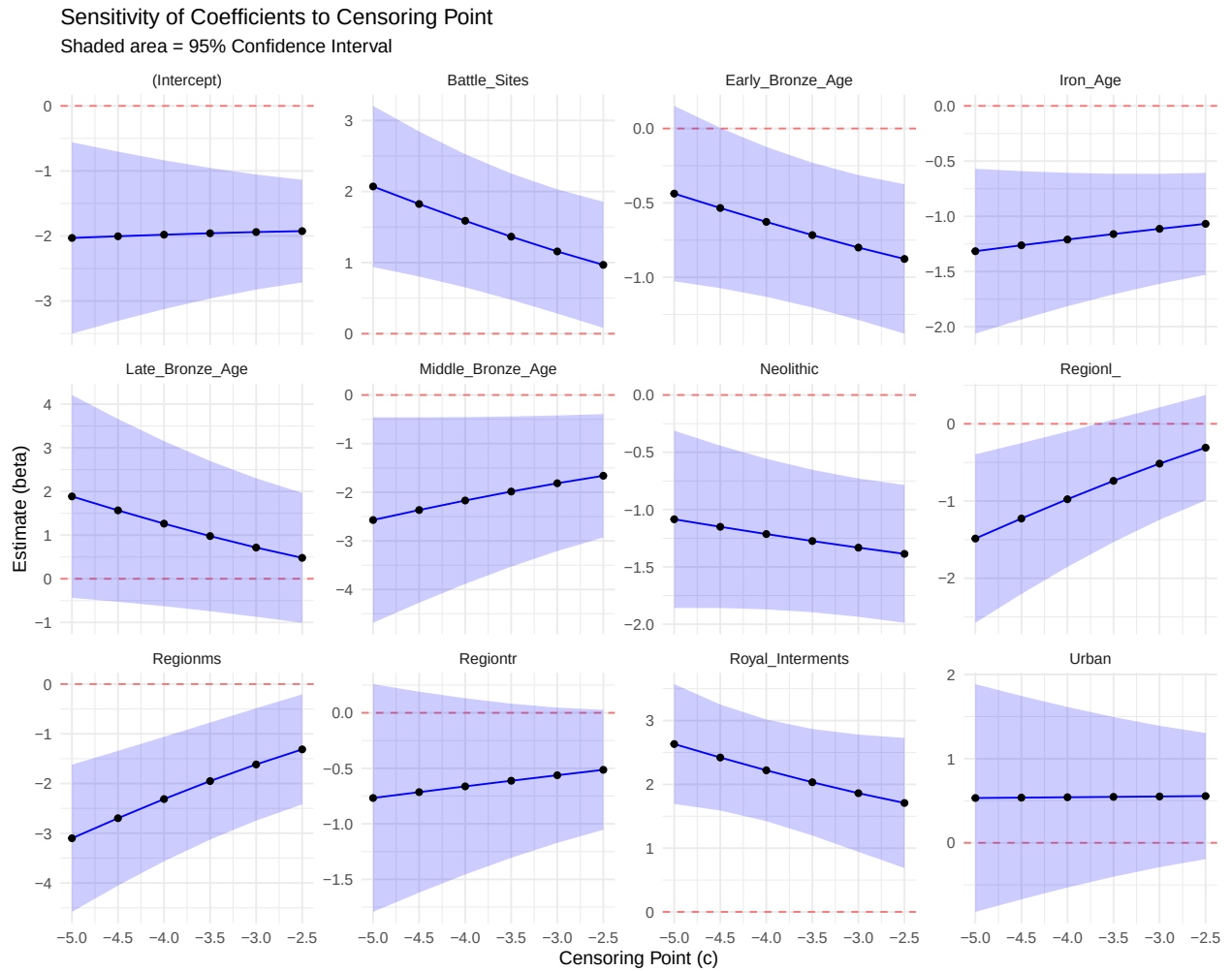


Figure 1: Sensitivity of estimated coefficients (beta) and 95% confidence intervals to the choice of censoring point c . The red dashed line indicates zero effect; intervals crossing this line indicate effects that are not statistically significant ($p > 0.05$).

Implication. The Tobit results—and in particular which period and proxy effects appear statistically supported—are sensitive to an analyst-chosen censoring threshold. This sensitivity reinforces that the Tobit/censoring formulation is not a robust basis for inference in a “count-out-of- n ” setting.

0.1.2 4.2 Comparing Model Fit and Predictive Accuracy to a Beta Binomial Specification

The analysis sample consists of $N = 82$ site-period observations (34 zero-trauma cases). To implement the beta-binomial model, continuous proportions were reconstructed into integer counts, ensuring the approximation is small (mean absolute discrepancy 0.021 individuals), so the reconstructed counts closely track the original shares.

While the Tobit model technically reports a lower AIC (234.7 vs 329.5), these metrics are **not comparable** because the models operate on different outcome scales (continuous log-density vs. discrete counts). Instead, we rely on “common-scale” predictive checks, which reveal severe miscalibration in the Tobit specification:

- **Predictive Validity (The “Exploding Mean” Failure):** The Tobit model generates predictions outside the possible range $[0,1]$ when projected back to the original scale. Because the latent normal is unbounded, $\exp(y^*)$ can be arbitrarily large; with heteroskedasticity from weights this yields extreme means. This is a property of exponentiating an unbounded normal index (lognormal moments), not a feature of the bounded data. The weighted predicted mean trauma share for the Tobit model was **2,330.16**, compared to an observed weighted mean of **0.091**. This failure is systematic across periods: for example, the Tobit model predicts weighted means of **5,677** for period `Early_Bronze_Age` and **7,896** for `Middle_Bronze_Age`, against observed means of < 0.10 . This occurs because the Tobit model applies analytic weights to the *log-scale* variance; when back-transformed to the natural scale, this inflated variance creates a heavy right tail that does not exist in the bounded $[0, 1]$ data.
- **Zero-Inflation/Censoring:** The Tobit model fundamentally misidentifies the frequency of zeros. It predicts a weighted zero rate of **59.9%**, whereas the observed weighted zero rate is only **12.6%**. The beta-binomial model successfully recovers the data structure with a predicted zero rate of **16.4%**.
- **Residual Diagnostics:** The beta-binomial model does not show evidence of misspecification. DHARMA simulation-based diagnostics confirm the residuals conform to expected distributions (KS test $p = 0.59$) and that dispersion is correctly modeled ($p = 0.36$), with no significant zero-inflation ($p = 0.58$).

0.2 5. Re-evaluating the Paper’s Conclusions

The choice of model alters the statistical significance of the findings, though not the direction of the effects (coefficient signs agree across all variables).

Table 2: Comparison of Statistical Significance ($p < 0.05$)

Variable	Context	Tobit (Original)	Beta-Binomial (Corrected)	Assessment
Battle_Sites	Violence	Sig ($p = .009$)	Sig ($p = .002$)	Robust
Royal_Interments	Status	Sig ($p < .001$)	Sig ($p < .001$)	Robust
Middle_Bronze_Age	Period 3	Sig ($p = .011$)	Sig ($p = .040$)	Robust
Neolithic	Period 1	Sig ($p < .001$)	NS ($p = .521$)	Not Supported
Early_Bronze_Age	Period 2	Sig ($p = .001$)	NS ($p = .854$)	Not Supported
Iron_Age	Period 5	Sig ($p < .001$)	NS ($p = .663$)	Not Supported
Regionms	Geography (Mesopotamia)	Sig ($p = .005$)	Borderline ($p = .051$)	Borderline Not Supported
Regionl	Geography (Levant)	NS ($p = .165$)	Sig ($p = .030$)	New Finding
Regiontr	Geography (Turkey)	NS ($p = .070$)	NS ($p = .653$)	Consistently NS

Note: P-values use cluster-robust standard errors with 21 clusters. Cluster-robust inference can differ substantially from model-based SEs for some coefficients. We apply the same cluster-robust/small-sample correction approach used for our Tobit replication to ensure maximal comparability across models.

A. Conclusions Supported by Both Models (Robust): * **Violence & Status:** Both models agree that **Battle_Sites** and **Royal_Interments** contexts are associated with significantly higher trauma prevalence. * **Period Middle_Bronze_Age:** Both models identify the **Middle_Bronze_Age** period as having significantly lower trauma than the baseline.

B. Conclusions Not Supported: The Tobit model identified three other periods (**Neolithic**, **Early_Bronze_Age**, **Iron_Age**) as having significantly lower trauma rates. **The beta-binomial model finds no evidence for this**, suggesting these effects could be artifacts of treating small-sample zeros as high-certainty data points.

C. New Insights: * **Geographic controls are model-dependent:** In the beta-binomial model **Regionl** (Levant) becomes statistically significant ($p = 0.030$) while **Regionms** (Mesopotamia) is borderline non-significant ($p = 0.051$), whereas the Tobit model supports **Regionms** ($p = 0.005$) but not **Regionl** ($p = 0.165$). Because geography is not the paper’s main focus, we treat these as secondary and emphasize the stability (or lack thereof) of the period and context effects.

Discussion of Robustness: Comparing the beta-binomial results to the Tobit threshold sensitivity analysis (Section 4.1) suggests two patterns:

1. **Tobit-sensitive terms also differ most across model classes** (**Early_Bronze_Age**, **Regionl**): These covariates show noticeable changes in support across plausible Tobit censoring points, and they also change when moving from the Tobit (paper implementation)

to the beta-binomial. This reinforces that inference for these terms is particularly dependent on the Tobit/censoring construction, whereas the beta-binomial avoids reliance on an analyst-chosen censoring threshold by modeling counts-out-of- n directly.

2. **Some Tobit-stable period effects are still not robust** (Neolithic, Iron_Age): Even when Neolithic and Iron_Age remain significant across Tobit censoring points, they are not supported in the beta-binomial model. This indicates that the difference is not only about the censor threshold, but also about deeper structural features handled by the count model—especially overdispersion and how information scales with sample size (n).

Summary: The robust evidence supports a simpler story: trauma prevalence is higher in Battle_Sites and Royal_Interments contexts and lower in Middle_Bronze_Age, while additional period differences (Neolithic, Early_Bronze_Age, Iron_Age) are not robust to a count model with overdispersion and appear sensitive to the Tobit/censoring specification.